Nikhil Rout

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NikhilRout   NikhilRout.github.io

Education

Bachelor of Technology, Electronics and Communication Engineering *2022 – 2026*
Vellore Institute of Technology, Chennai *CGPA: 9.15/10*

Research Experience

Visiting Research Assistant *Los Angeles / Remote*
Vortex GPGPU, ORCAS Lab, UCLA *May 2025 – Present*

- Developed a Tensor Core Unit (TCU) extension for the open-source RISC-V based Vortex GPGPU, accelerating WMMA operations in deep learning kernels
- Architected and implemented a configurable Microscaling (MX) Fused Dot Product datapath to explore custom block-floating-point formats
- Performed microarchitecture design space explorations, evaluating format-wise shared multipliers, Modulo-4 operand grouping CSA, and threads/warp arithmetic intensity scaling
- Introduced a novel sparse lane mask clock-gating strategy for dynamic power reduction in workloads exhibiting dual-side unstructured sparsity
- Validated TCU FP numerical accuracy by porting the FTTN benchmark to Vortex WMMA API
- Achieved 4-cycle latency at 300 MHz Fmax on the Xilinx U55C FPGA, delivering ∼3.1× throughput over the open-source baseline (Berkeley HardFloat) at less than 60% the area cost
- Demonstrated end-to-end speedups on Llama2-inference benchmarks: 9.2x (prefill) and 2.0x (decode)

Summer Research Intern *Chennai*
Centre for Nanoelectronics and VLSI Design, VIT Chennai *May 2024 – Jul 2024*

- Investigated radix-2 DIT and DIF butterfly structures to evaluate optimal dataflow for FFT and Number Theoretic Transform hardware accelerators
- Designed pipelined NTT/INTT cores with modulo-arithmetic operations for polynomial multiplication in lattice-based post-quantum cryptography schemes on FPGAs

Research Publications

[1] **Nikhil Rout**, Blaise Tine, "Ten-Four: An Open-Source Fused Dot Product Unit for Mixed-Precision GPGPU Tensor Cores". In Preprint, Under Review

[2] **Nikhil Rout**, Jean Jenifer Nesam J, "Optimizing RGB to Grayscale, Gaussian Blur and Sobel-Filter operations on FPGAs for reduced dynamic power consumption". In *IEEE Artificial Intelligence For Internet of Things (AIIoT)*, 2024.

Talks and Workshops

- Rethinking Tensor Core Microarchitecture for Emerging Microscaling Format Support on GPUs
  Workshop on Open-Source Computer Architecture Research (OSCAR) @ ISCA 2026
- Exploring Fused Dot Product Microarchitecture for the Vortex Tensor Core Unit Extension
  Vortex Workshop and Tutorial @ MICRO 2025

Skills and Competencies

Languages: Verilog/SystemVerilog, C/C++, OpenCL, CUDA, HIP, Python, PyTorch

Tools: Quartus Prime, Xilinx Vivado/Vitis, Synopsys VCS/DC, Make, Git, LaTeX

Simulators: Vortex SimX, Accel-Sim/GPGPU-Sim, gem5, Ramulator

Honors, Awards and Service

- Short-Term Research Scholar, UCLA CS ($27K)
- Student Volunteer, HotChips 2026
- Student Travel Grants - ISCA 2026 ($1000), MICRO 2025 ($580)
- Teaching Assistant, VIT Chennai (FPGA System Design 2025)
- Best Paper Award, IEEE AIIoT 2024

References

Dr. Blaise-Pascal Tine, *Assistant Professor, UCLA* blaisetine@cs.ucla.edu
Dr. Jean Jenifer Nesam J, *Assistant Professor, VIT Chennai* jeanjenifernesam.j@vit.ac.in